MME 386 Marine Electrical Technology and Electrical Measurements

Unit 2: **Electric Propulsion**

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Content

- Electric propulsion – systems, difference and relative merits
- Alternating current diesel electric and turbo electric propulsion system engines
- Generator excitation systems and control

Introduction

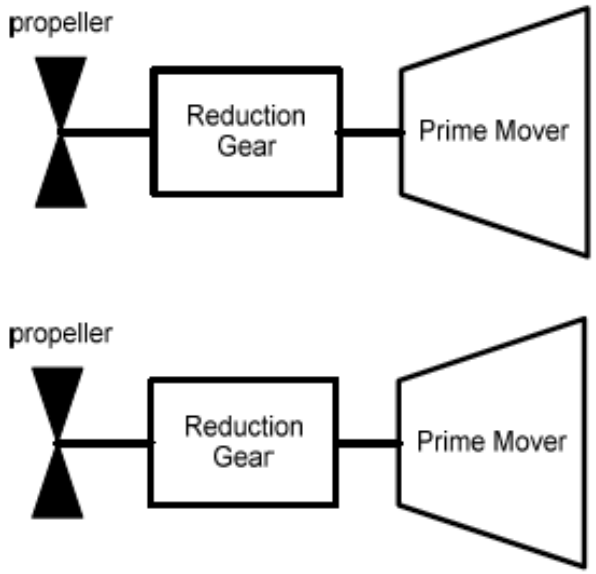
- Electric installations are present in any ship, from
  - powering of communication and navigation equipment,
  - alarm and monitoring system,
  - running of motors for pumps,
  - fans or winches,
  - to high power installation for electric propulsion.

Introduction

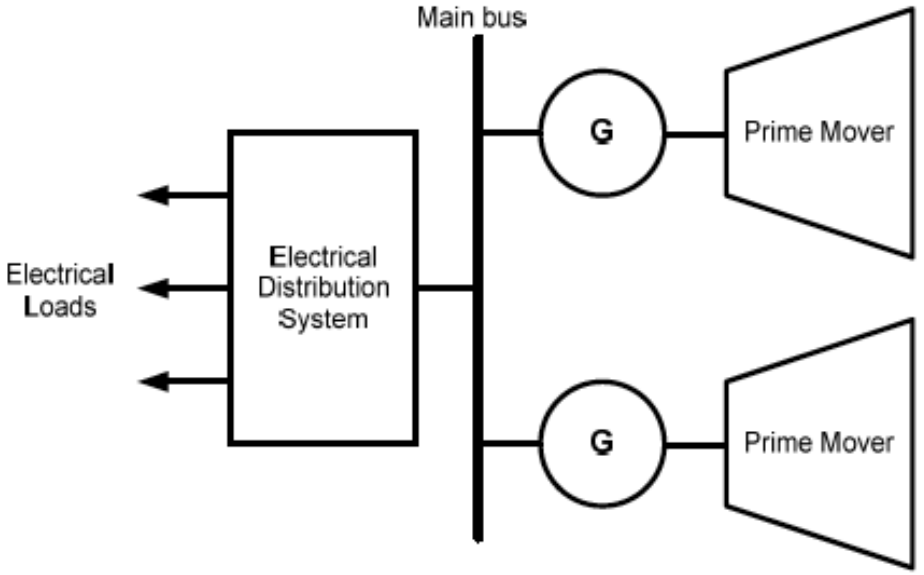
- Electric propulsion is an emerging area where various competence areas meet.
- The objective of this unit is to give an introduction to installations for electric propulsion.

Segregated Power System

- Prime Movers
  - diesel or gas turbine
- Propulsion
  - consumes 80-90% of total power
- Generators
  - Coupled to same prime movers

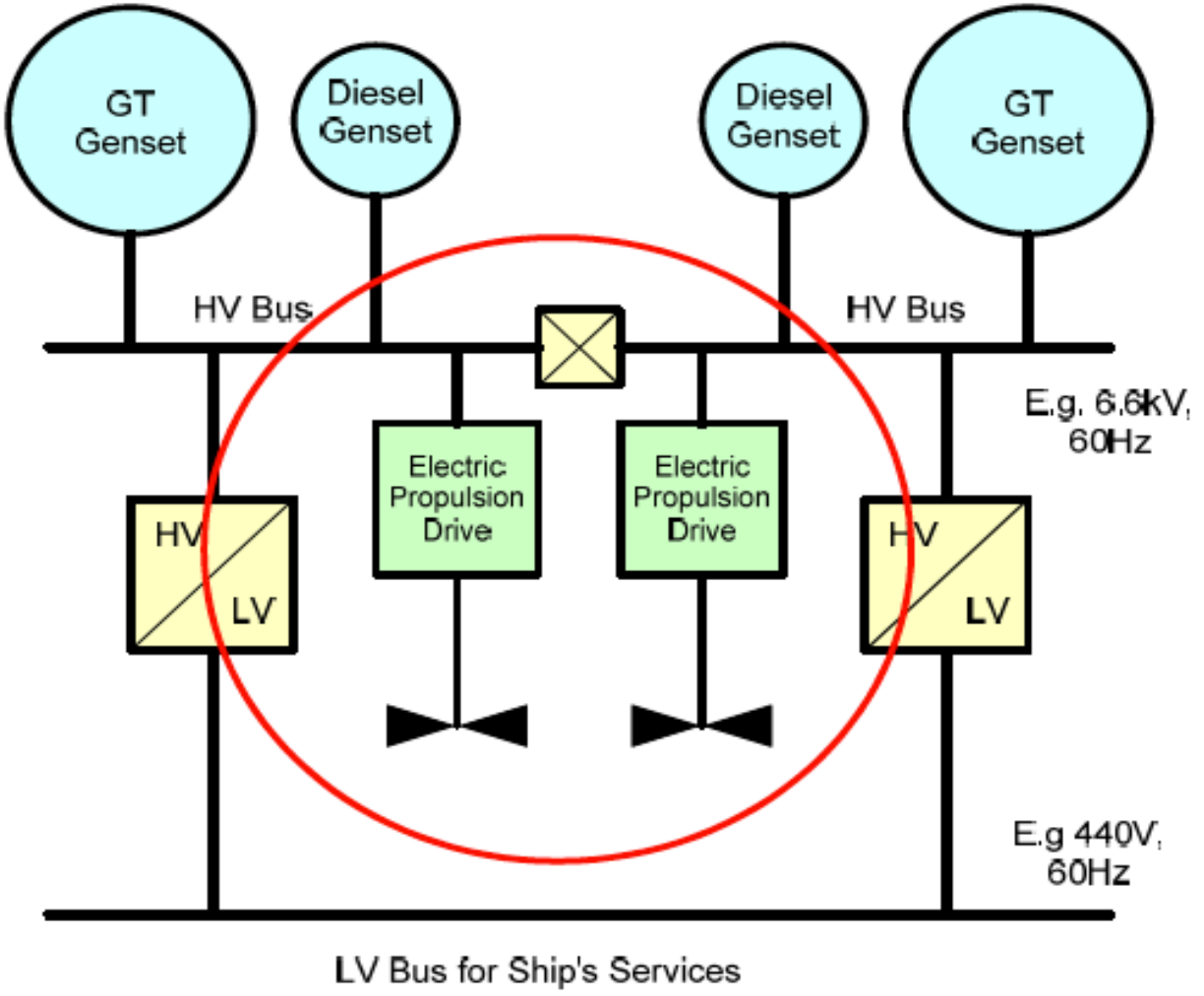


Mechanical Propulsion System



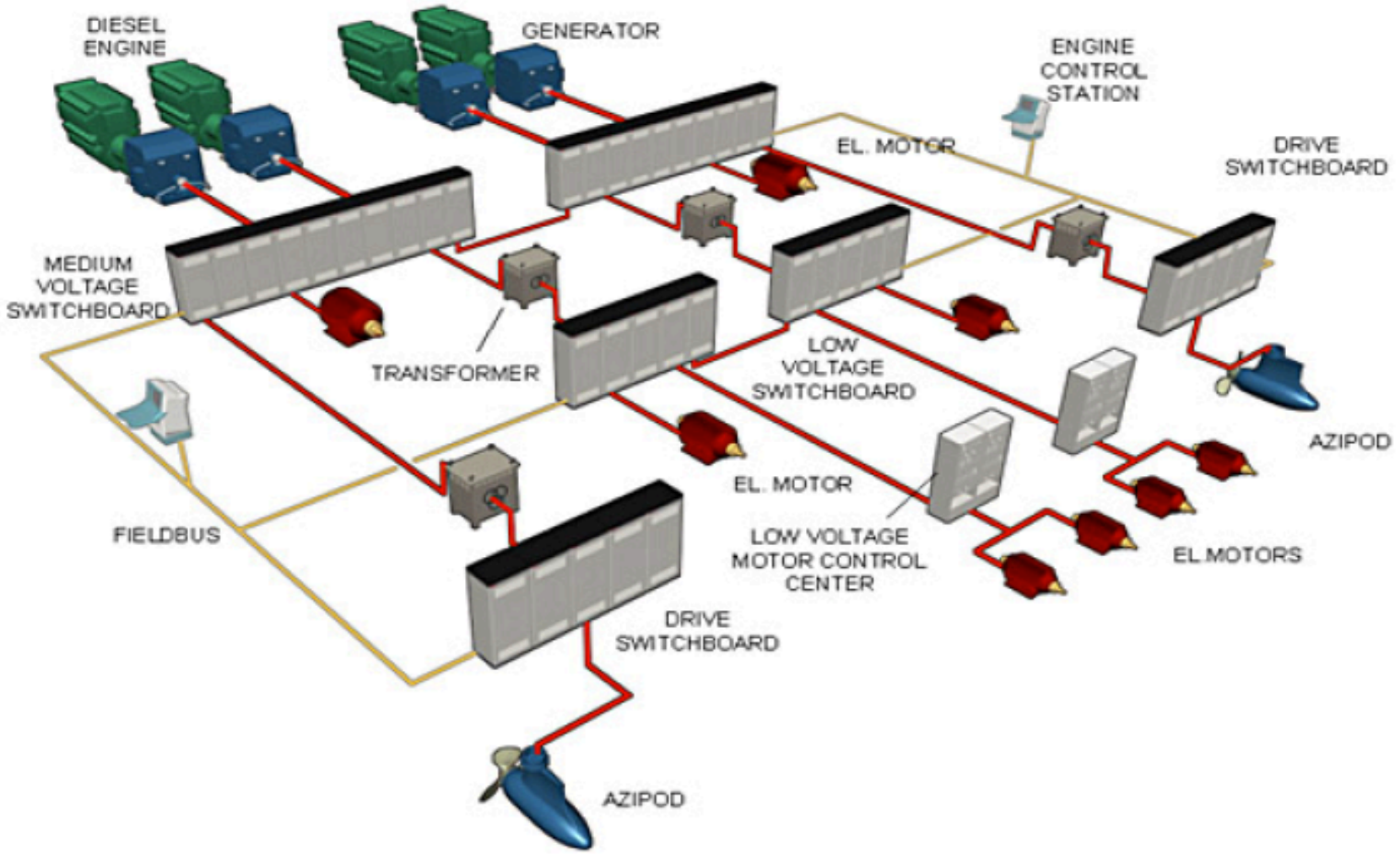
Electrical Generation and Distribution

Integrated Power System



Mixture of diesel and Gas Turbine to optimise power and efficiency for different operating conditions

Integrated Power System



Integrated Power System

- Common set of generators to power
  - Electric propulsion systems
  - Ships services (hotel loads)
  - Combatant loads

Electric Propulsion

- The concept of electric propulsion is not new, the idea originated more than 100 years ago.
  - Submarines – a relatively small vessel
- Advances in technologies for variable speed drives (DC and then AC) has led to the development of efficient multi-megawatt drives.



Electric Propulsion

- **Advantages**
  - Reduced fuel consumptions (esp. if there is a large variation in load) and maintenance
    - Mixed prime movers (gas turbine/diesel)
    - Remove gearbox
  - Controllability, fast response and better maneuvering
  - Distribution of weight throughout the ship
  - Reduction of propulsion noise and vibrations
  - Reduction of volume – especially with podded systems
- **Disadvantages**
  - Increased investment costs
  - Bigger variety of component
    - Service skills and spare parts

Maneuverability



Electric Propulsion Drive



System Efficiency

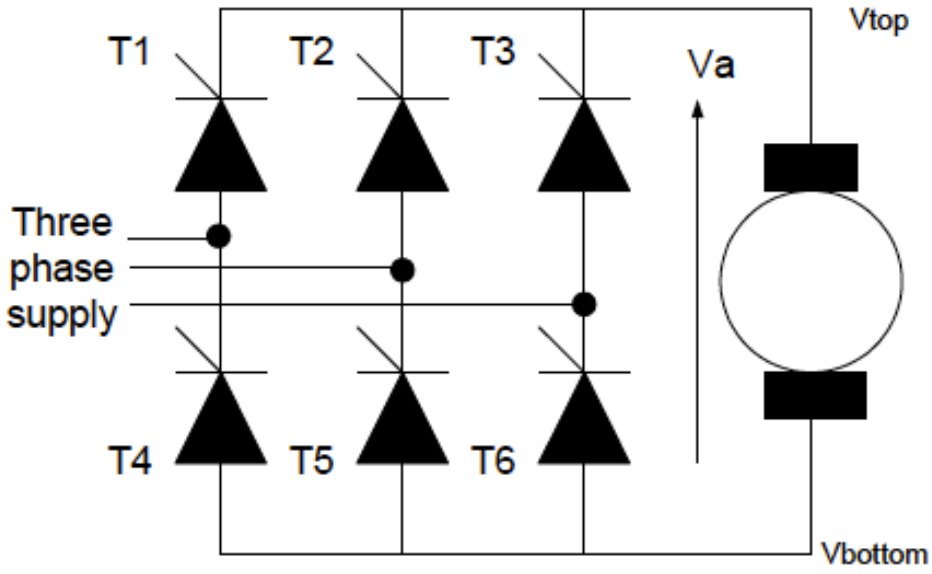
- Generator – 95 – 97%
  - Switchboard – 99.9%
    - Transformer – 99-99.5%
      - Power Electronic Converter – 98-99%
        - Motor – 95-97%
- Overall Efficiency – 88 – 92% at full load excluding prime mover
- Efficiency savings made through independent variable speed operation of the diesel engine and the propeller

Electric Propulsion Drives

- Initially variable speed DC drives
  - Controlled Rectifier (thyristors)
- Variable speed AC drives introduced in the 1980s
  - Higher power to weight
  - Lower maintenance (no commutator)
  - Cycloconverter
  - Current Source Inverter (CSI)
  - Voltage Source Inverter (VSI)

DC Drives

- AC supply – fixed frequency and amplitude
- Control the firing angle of a fully controlled thyristor rectifier to vary armature voltage (and speed)



DC Drives

- Thyristors – relatively cheap
- Control – straightforward and fast

- Poor motor waveforms
- Poor supply displacement factor at low speed
- Significant harmonic current of supply currents

- Practical limit ~ 2-3 MW
  - Armature current needs to pass through a commutator and brushes

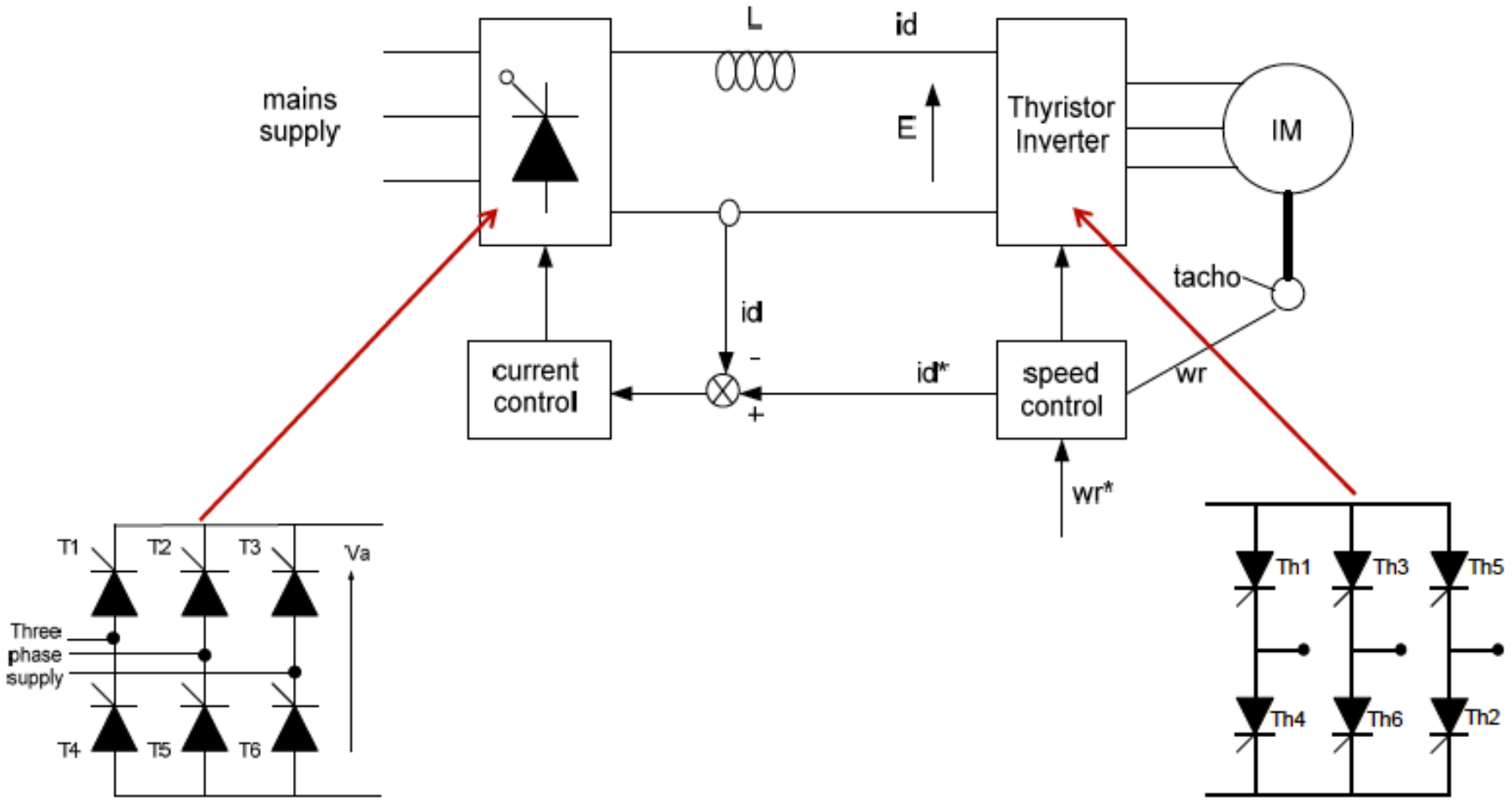
AC Drives

- Current Source Inverter (CSI) Drives
- Cycloconverter Drives
- Voltage Source Inverter (VSI) Drives

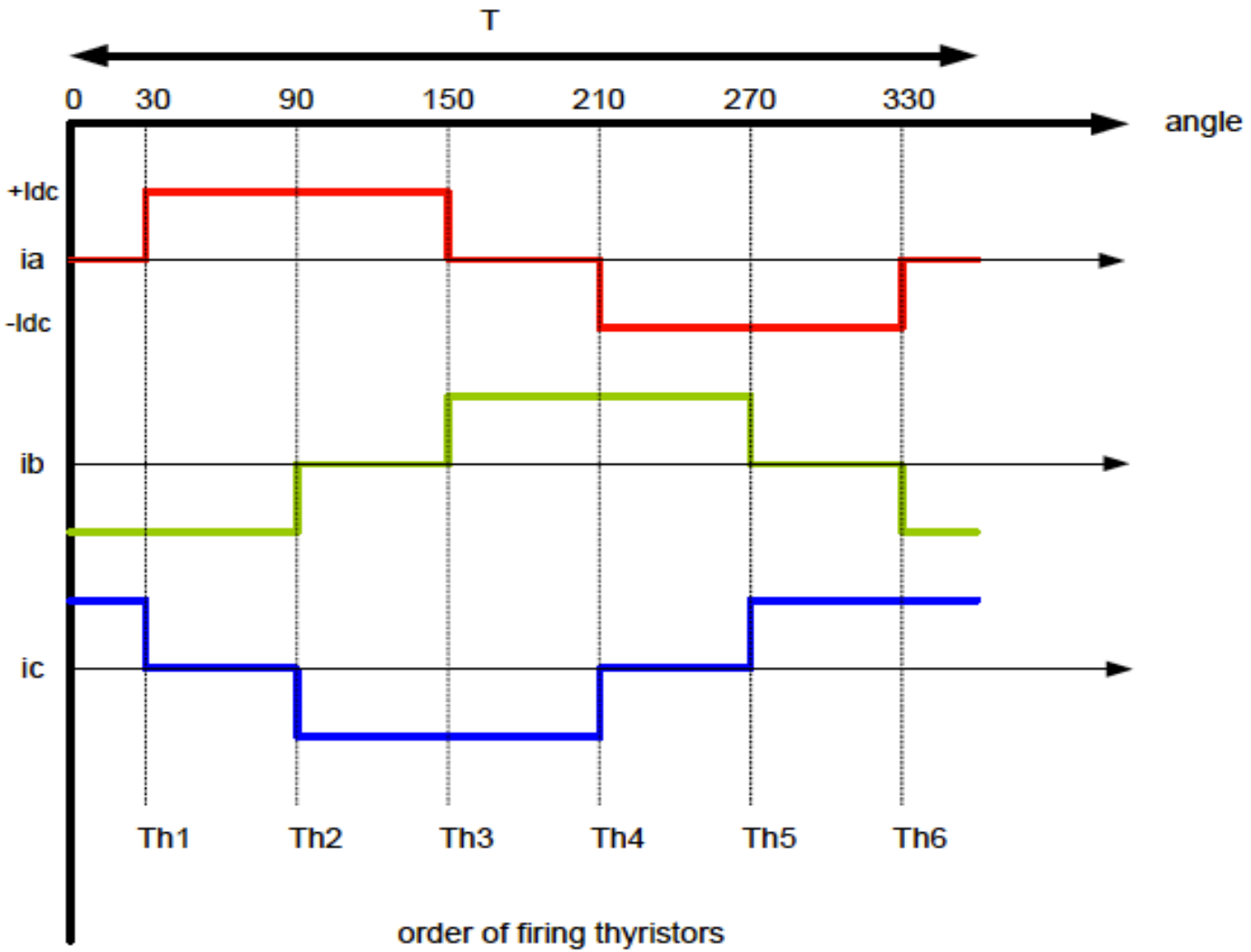
Current Source Inverter

- AC motor drive using thyristor technology
- Used for induction motors and (more commonly)
  - synchronous AC motors
- Sometimes six stator windings fed from two CSIs
- Supply interface – Controlled rectifier feeding a large inductor
  - Controls DC link current
- Inverter directs DC current into required phases of the AC machines
- Up to 100 MW!

Current Source Inverter

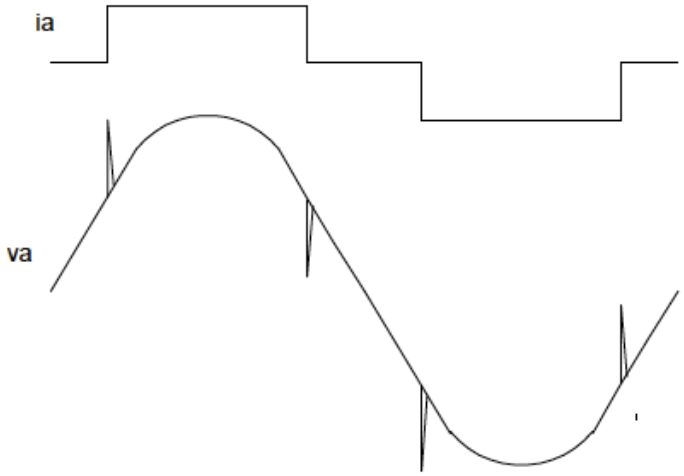


Current Source Inverter



Current Source Inverter

- At low shaft speeds the torque pulsations may be objectionable and PWM methods may have to be employed.
- PWM cannot be used across the whole frequency range due to switching frequency limitations.
- Stator voltage will probably be sinusoidal, but will contain spikes due to the effect of $l_s \frac{di_s}{dt}$.



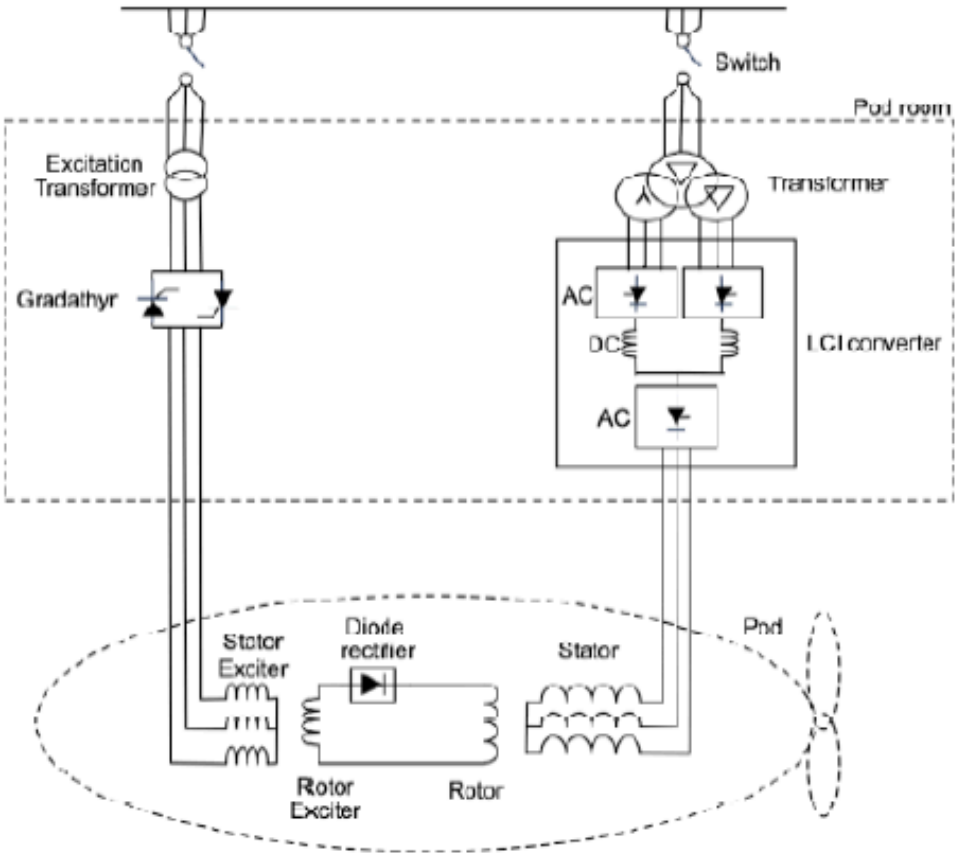
CSI Application

- Cruise Liner with 2 pods
- Each pod contains a 19 MW synchronous motor. Each motor is fed from a current source inverter mounted in the main body of the ship

Rolls Royce and Alstom



CSI Application



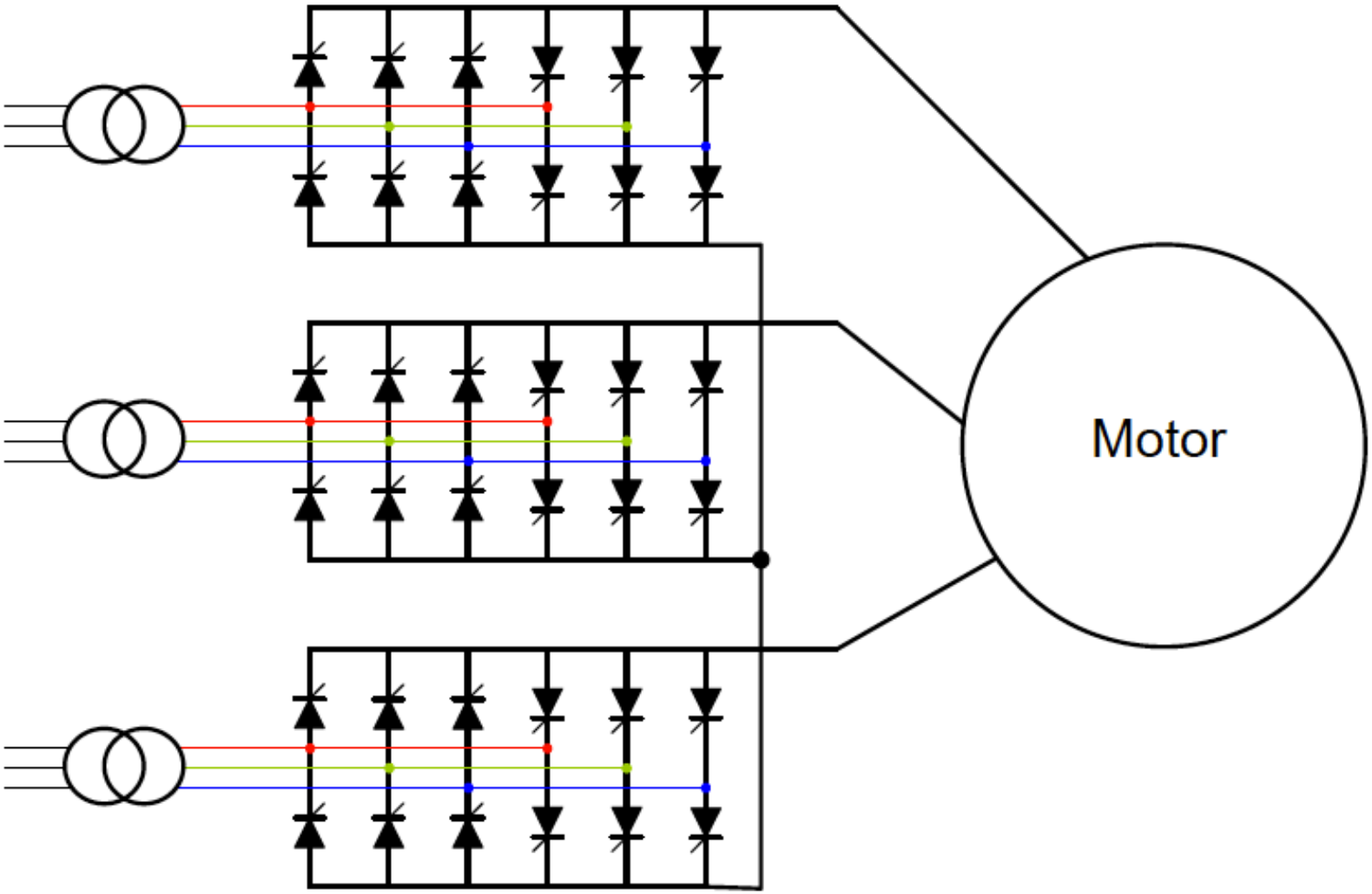
Circuit Diagram for the Rolls Royce Mermaid Pod Propulsion system

L Bergh and U Hellden, "Electrical systems in pod propulsion" MSc Thesis Chalmers University of Technology, Sweden, 2007

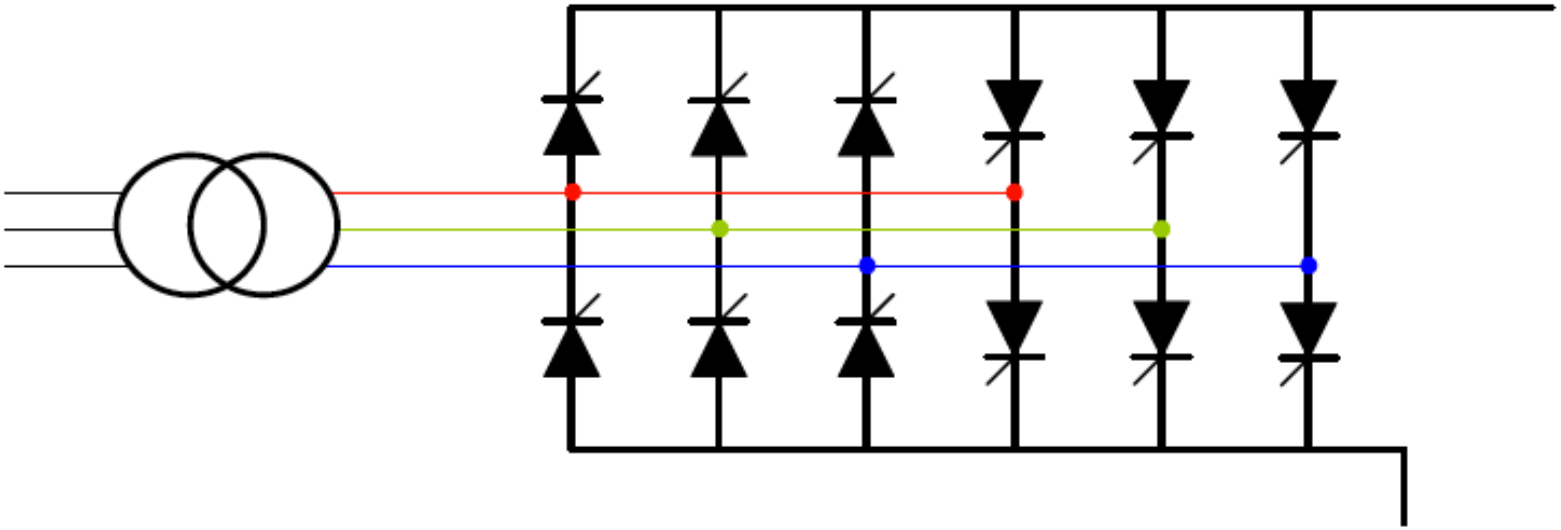
Cycloconverter

- A high power direct AC-AC Converter
- Output voltage synthesized by switching parts of the input voltage directly onto the motor using thyristor bridges
- Can control the motor frequency to about 1/3 the supply frequency
  - Good for low speed performance eg ice breakers
- Better motor current waveform than CSI
- 2-25 MW

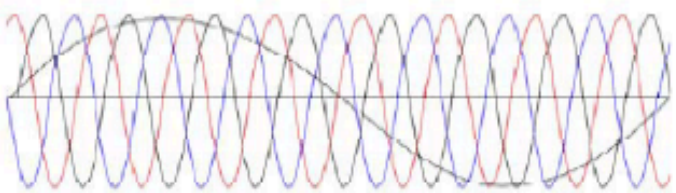
Cycloconverter

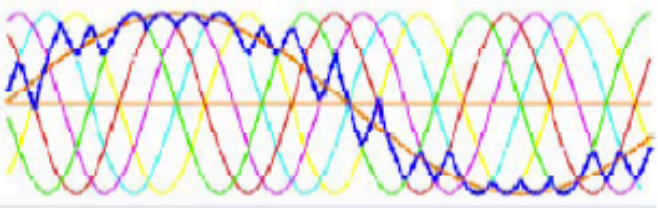


Cycloconverter – one phase

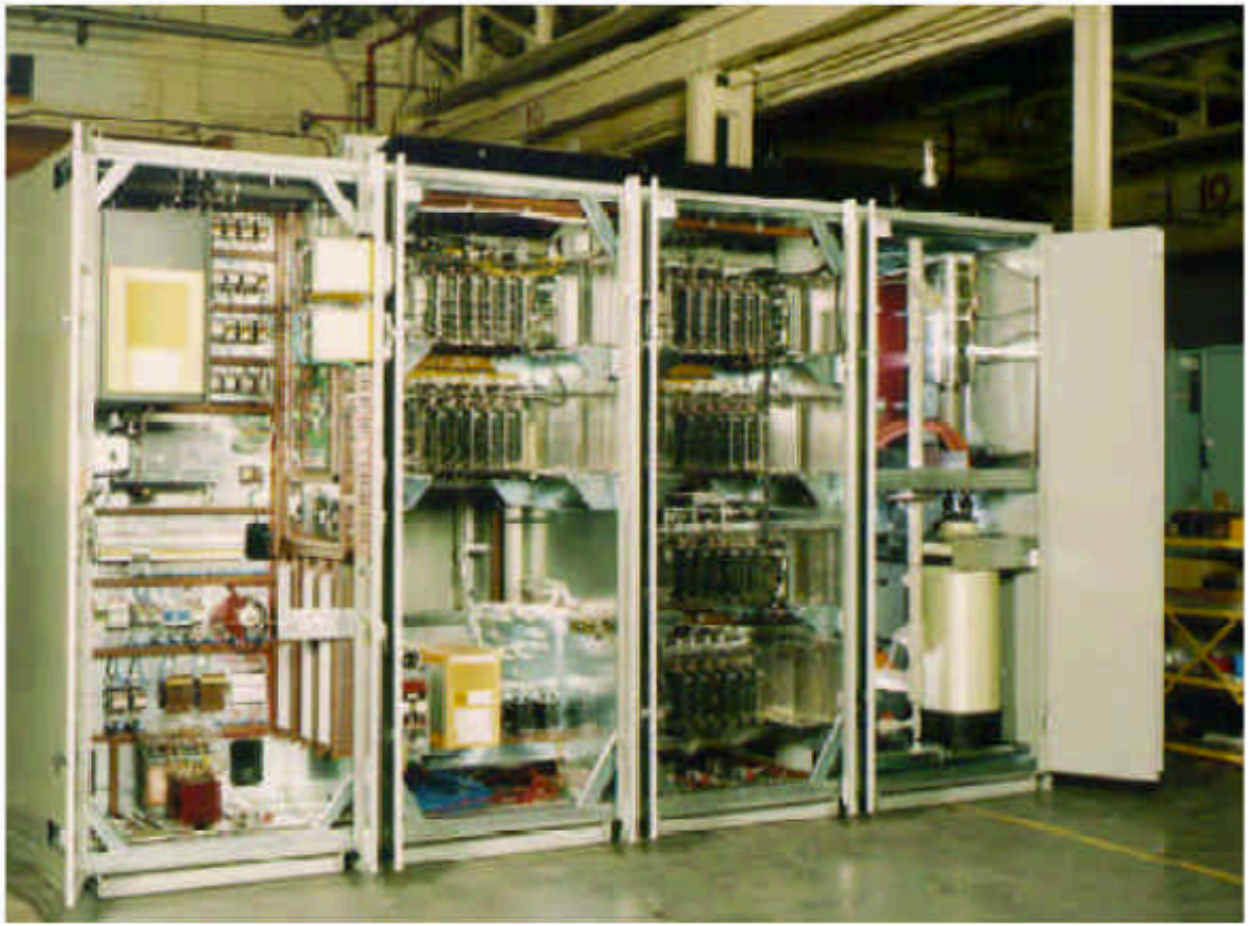


- Each phase must be isolated from the others using a supply side transformer
- 2 bridges per phase – bidirectional power flow
- Vary the firing angle α as a sinusoidal function of desired output frequency





Cycloconverter



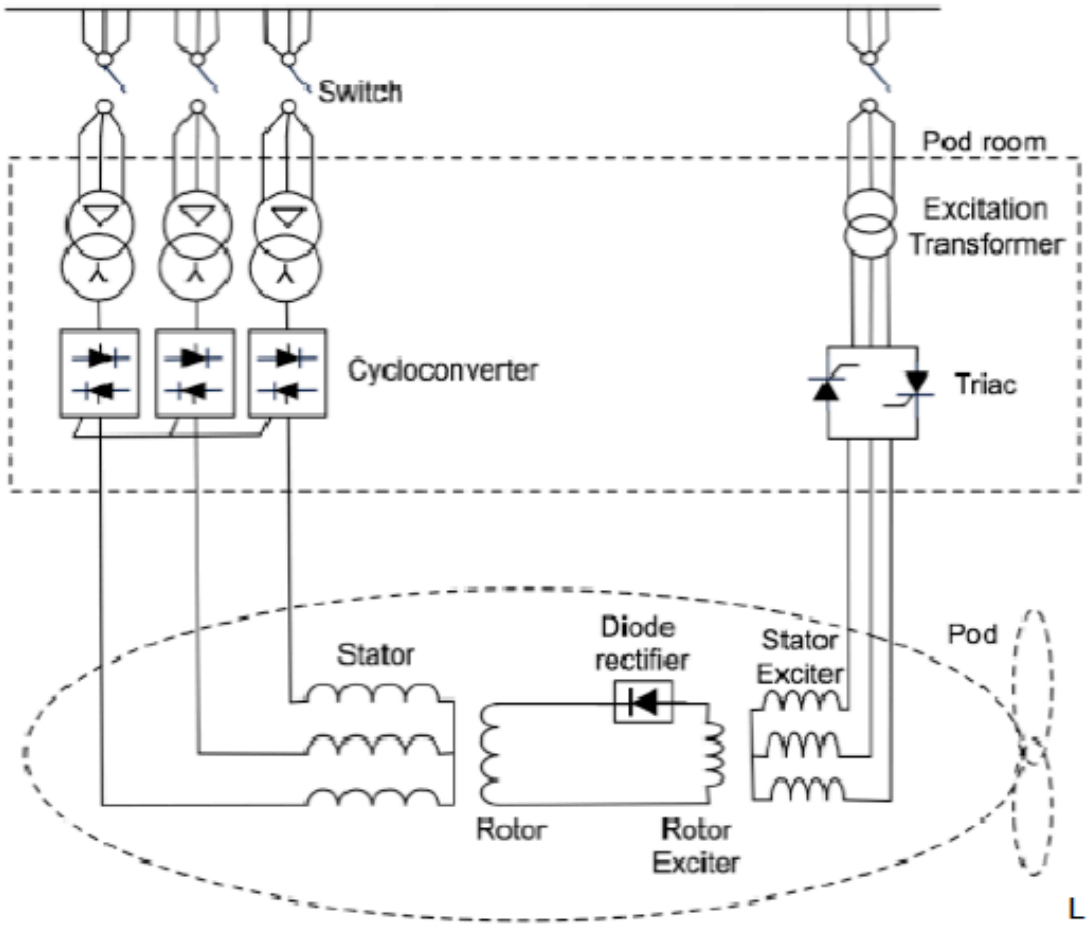
5.6 MW water cooled cycloconverter, used in Auxiliary Oiler and Landing Platform Dock – *Hodge and Mattick, "The Electric warship, then Now and later", INEC 2008, Hamburg*

Cycloconverter Application



- Cycloconverter drive applied to an icebreaker
- Large power at low speed, and high dynamic performance
- Two shafts, each powered by two 5.6 MW Cycloconverters

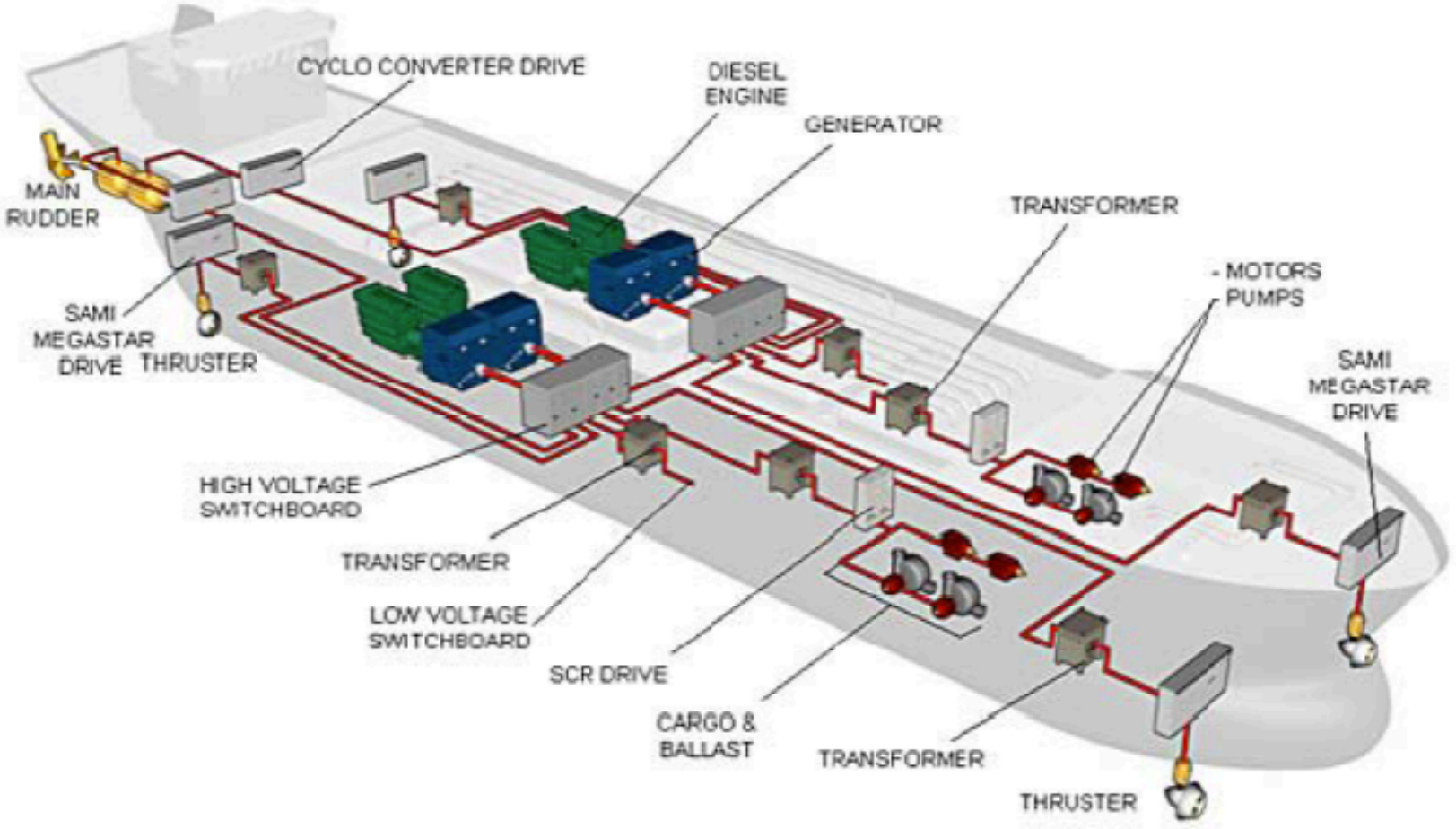
Cycloconverter Application



Circuit diagram of ABB pod propulsion unit "Azipod"

L Bergh and U Hellden, "Electrical systems in pod propulsion" MSc Thesis Chalmers University of Technology, Sweden, 2007

Cycloconverter Application



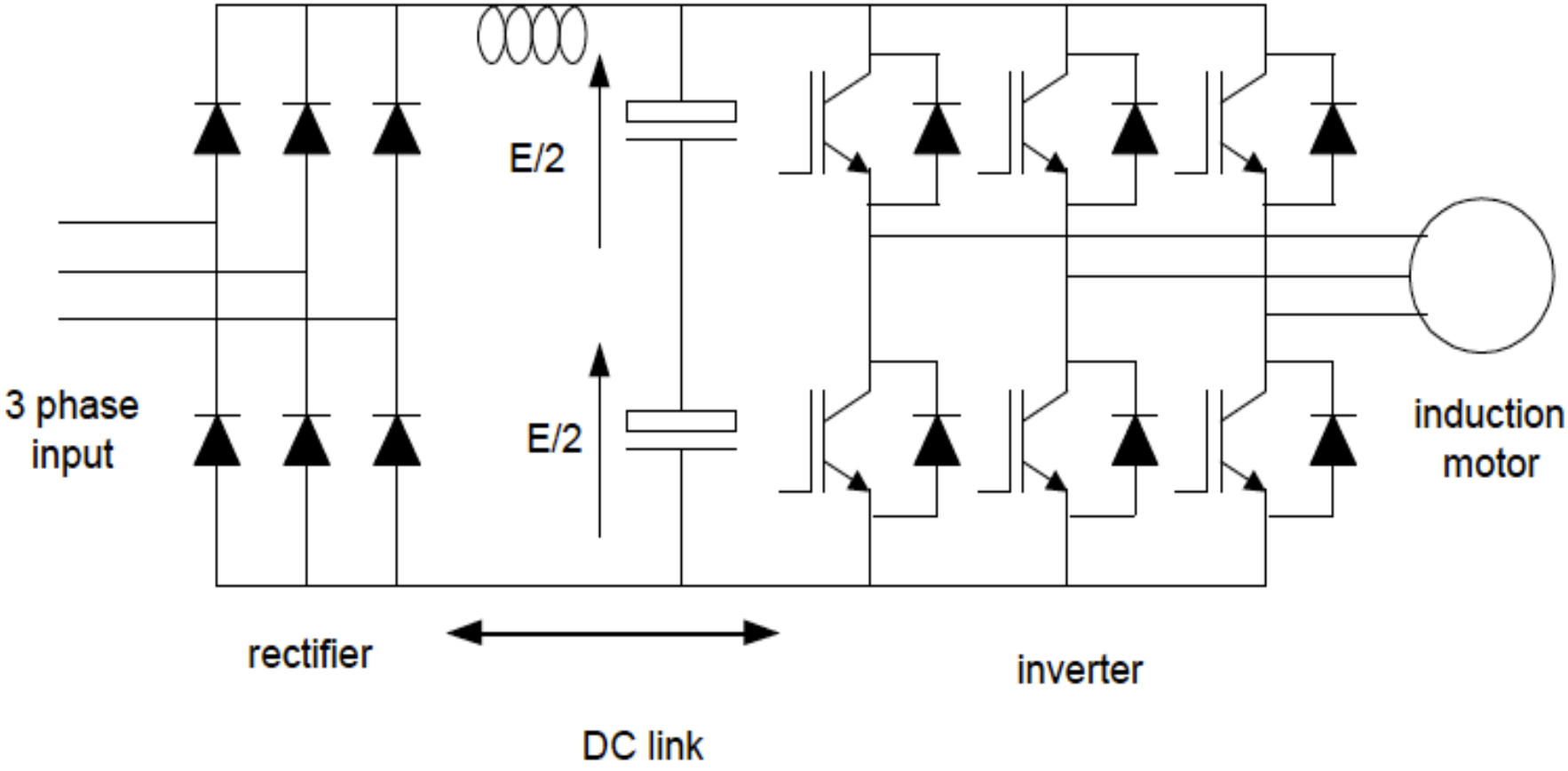
Shuttle Tanker with cycloconverter propeller drives and DC drives for cargo and ballast pumps – *ABB website*

Voltage Source Inverter

- Most modern variable speed drives employ VSI technology
  - Fast control (vector control)
  - Good motor harmonic currents – low torque pulsations and losses
  - Induction motor or permanent magnet synchronous motor
- Output voltage controlled using Pulse-width Modulation (PWM)
- Input current harmonics poor
  - Multi-pulse techniques
  - Filters on the supply side

Voltage Source Inverter

- Basic Topology



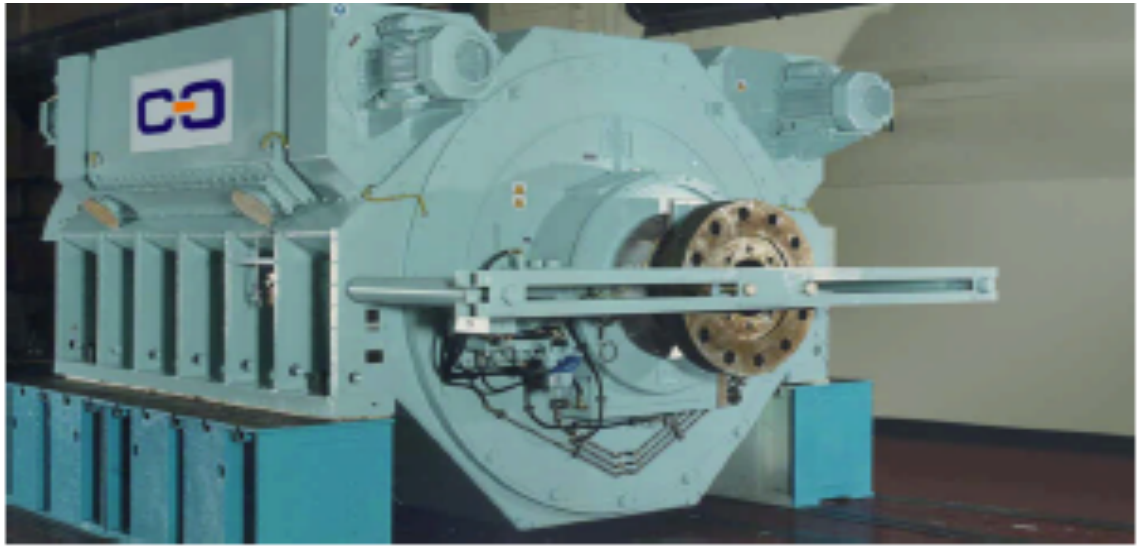
Recent Advances in Drive Systems

- The Type 45 Destroyer is the first use of a VSI for propulsion of a warship
- The system uses an H-bridge for each individual phase of the Advanced Induction Motor (AIM)
- 15 different H-bridges arranged as 3 sets of five phase groups



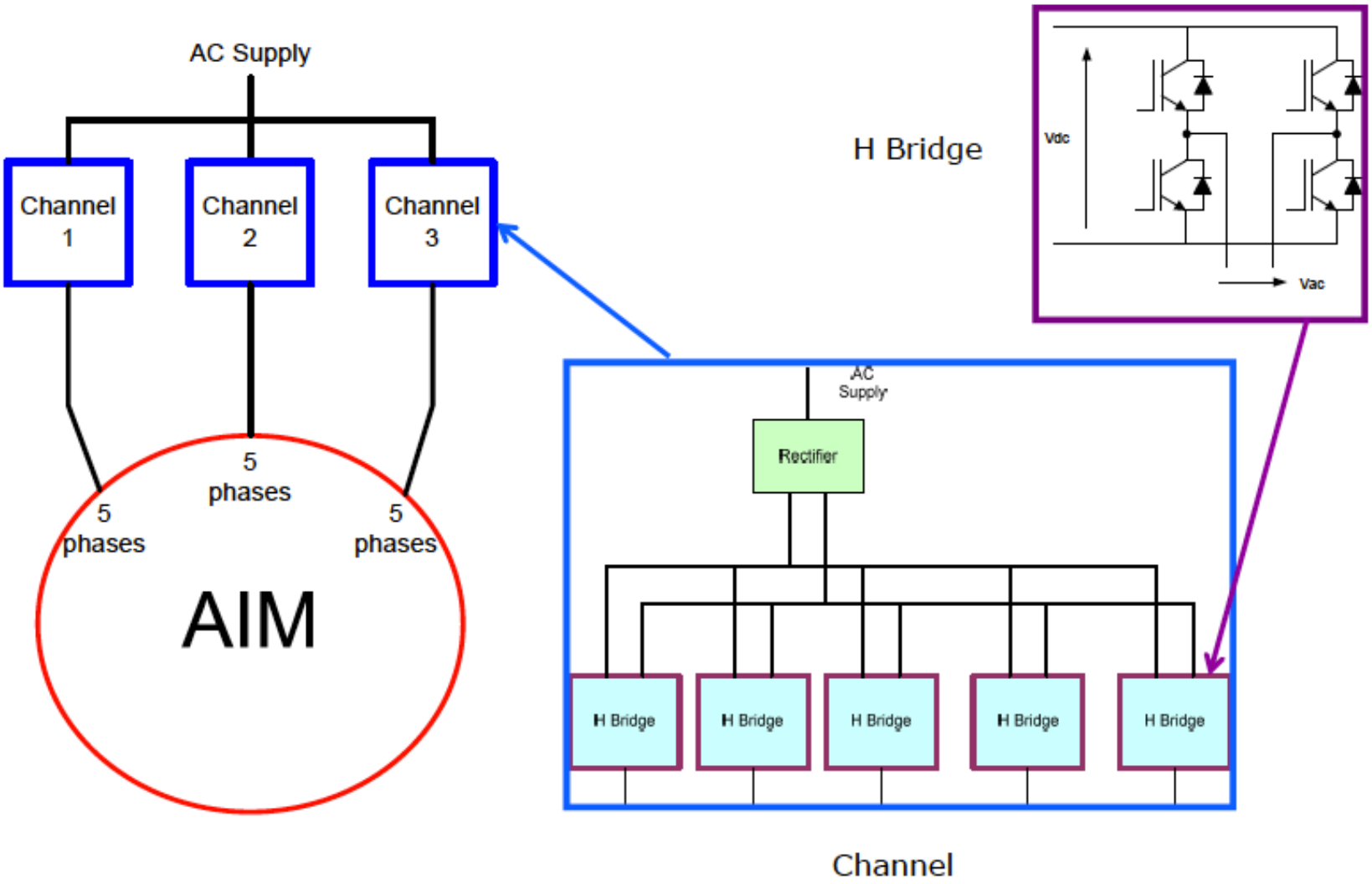
Recent Advances in Drive Systems

- Advanced Induction Motor
  - Fault tolerance (15 phase)
  - Power density – more current into the machine



  - 15 phase
  - 20 MW @ 180 rpm
  - 89 tonnes (with ancillaries)

VSI Drive for AIM



Generator Excitation Systems

- The majority of new buildings and all commercial vessels have an AC power generation plant with AC distribution.
- The generators are synchronous machines, with a magnetizing winding on the rotor carrying a DC current,
- And a three-phase stator winding where the magnetic field from the rotor current induces a three-phase sinusoidal voltage when the rotor is rotated by the prime mover.

Generator Excitation Systems

- The frequency f [Hz] of the induced voltages is proportional to the rotational speed N_s [RPM] and the pole number p in the synchronous machine:

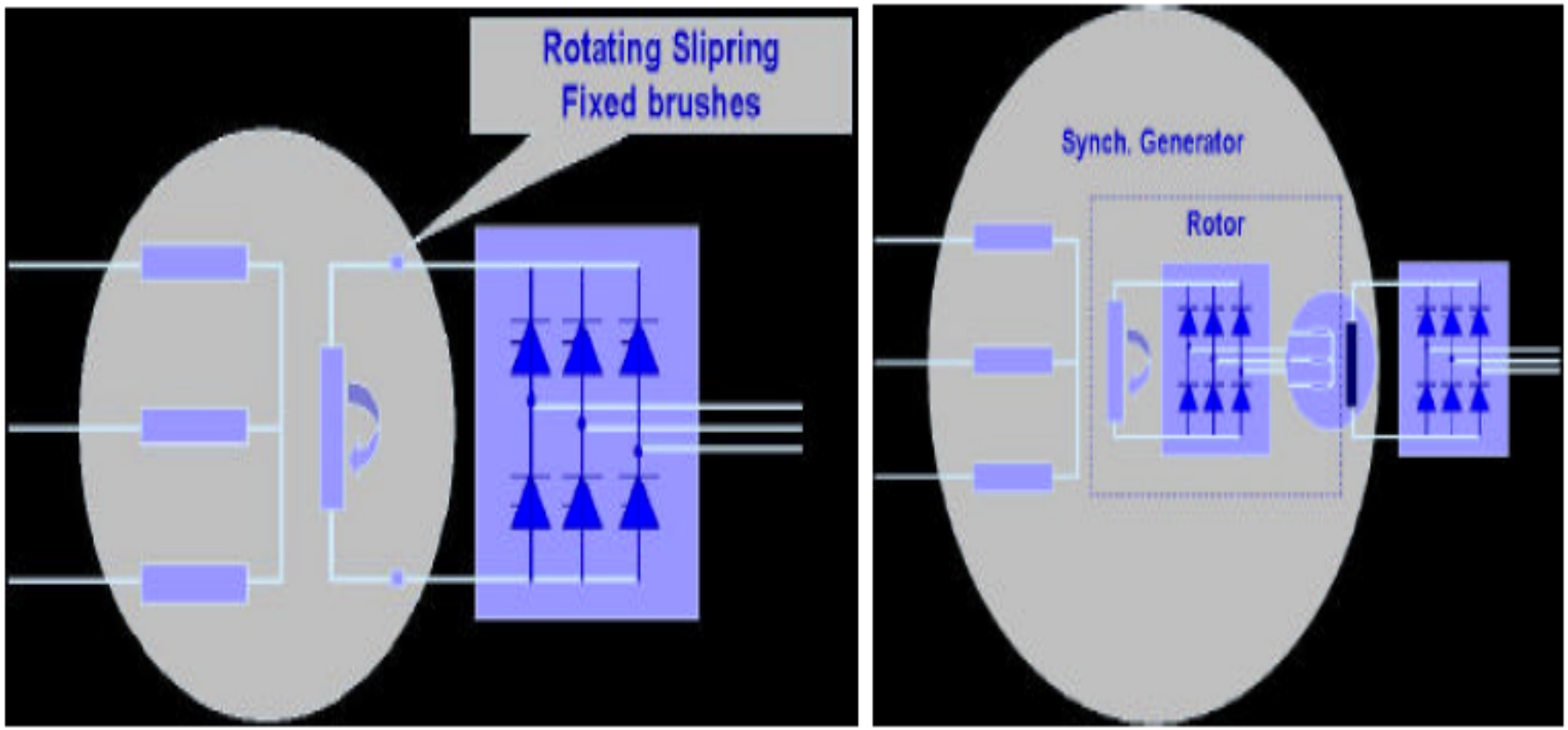
$$f = \frac{p}{2} \cdot \frac{N_s}{60}$$

- The DC current was earlier transferred to the magnetizing windings on the rotor by brushes and slip rings.

Generator Excitation Systems

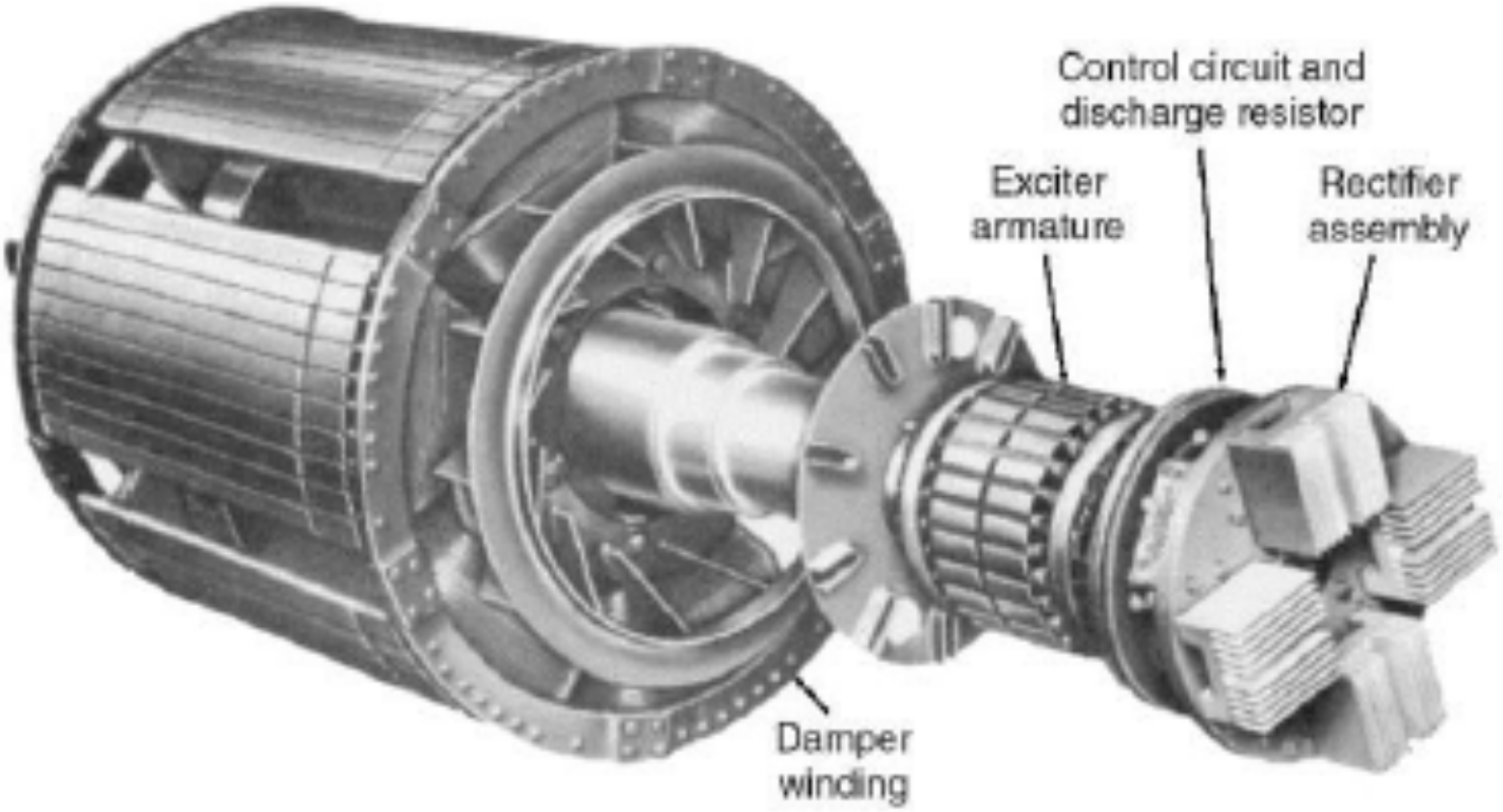
- Modern generators are equipped with brushless excitation for reduced maintenance and downtime.
- Modern brushless synchronous generators can either have:
  - Permanent magnet on rotor or
  - Exciter plus rotating rectifier
- For the case of exciter plus rotating rectifier, the rectified current then feeds the field windings.

Generator Excitation Systems



Magnetization of the rotor winding, left: brushed; right: brush-less magnetization

Generator Excitation Systems



Three phase exciter rotor to avoid slip rings and get dc power to field

Generator Excitation Systems

- The excitation is controlled by an automatic voltage regulator (AVR), which senses the terminal voltage of the generator and compares it with a reference value.
- Simplified, the controller has PID characteristics, with stationary limited integration effect that gives a voltage drop depending on the load of the generator.

Generator Excitation Systems

- The voltage drop ensures equal distribution of reactive power in parallel-connected generators. According to most applicable regulations, the stationary voltage variation on the generator terminals shall not exceed ±2.5% of nominal voltage.
- Also, the largest transient load variation shall not give voltage variation exceeding -15% or +20% of the nominal voltage unless other has been specified and accounted for in the overall system design.
- In order to obtain this transient requirement, the AVR is normally also equipped with a feed-forward control function based on measuring the stator current.

Generator Excitation Systems

- In addition to the magnetizing winding, the rotor is also equipped with a damper winding which consists of axial copper bars threaded through the outer periphery of the rotor poles, and short circuited by a copper ring in both ends.
- The main purpose of this winding is to introduce an electromagnetic damping to the stator and rotor dynamics.
- A synchronous machine without damper winding is inherently without damping and would give large oscillations in frequency and load sharing for any variation in the load.

END